

M2 Quantitative Finance
Stochastic Control

Optimal Dividend and Capital Structure with
Debt Covenants

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Abstract

We study an optimal dividend and recapitalization problem for a firm subject to debt and financial covenants. In contrast to standard frameworks where default immediately triggers liquidation, the model distinguishes between financial distress, technical default, and liquidation. When a credit event occurs during distress, the firm enters default and is granted a finite grace period during which capital injections may restore solvency. Liquidation occurs only if recovery fails within this period or if assets reach zero. The firm aims to maximize the expected discounted value of dividends net of capital injections up to liquidation. The problem leads to a singular stochastic control framework with regime switching and endogenous liquidation time, characterized by a coupled system of Hamilton–Jacobi–Bellman variational inequalities. We analyze the structure of the value functions, discuss the associated optimal regions, and provide numerical illustrations of the resulting policies. We further compare our numerical output with the published illustrations and document a systematic discrepancy in the fixed-point value $a^* = v_0(D)$, discussing its possible sources.

Keywords: Optimal dividend policy; Capital structure; Debt covenants; Singular stochastic control; Regime switching; Bankruptcy; Variational inequalities.

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1 Introduction

Dividend policy and capital structure decisions are central issues in corporate finance, as they jointly determine shareholders' payouts, firms' solvency, and long-term survival. A large body of literature has studied these problems under uncertainty using continuous-time stochastic models. However, while the mathematical tools employed across these contributions are often similar, the economic interpretation of default differs significantly. In particular, most models treat default as an absorbing event that immediately triggers liquidation, whereas in practice financial distress may involve an intermediate phase.

A dominant assumption in both structural credit risk models and stochastic control models of corporate policies is that default coincides with immediate liquidation. In the structural approach initiated by [Merton \(1974\)](#), the firm's asset value follows a diffusion process and default occurs when assets fall below the face value of debt. Extensions by [Black and Cox \(1976\)](#) introduce early default through a barrier mechanism. In these frameworks, once the asset value crosses the default threshold, the firm is effectively liquidated, and the analysis focuses on the valuation of contingent claims.

A similar assumption appears in the literature on optimal dividend policies. The seminal works of [Jeanblanc-Picqué and Shiryaev \(1995\)](#) model the firm's cash reserves as a controlled diffusion process, where dividend payments are represented through a non-decreasing control. Bankruptcy occurs when reserves reach zero, and liquidation follows immediately. Subsequent extensions, including those by [Asmussen and Taksar \(1997\)](#), maintain this absorbing boundary structure.

More recent models incorporate investment and financing decisions. For instance, [Løkka and Zervos \(2008\)](#) analyze interactions between dividend payments and equity issuance, while [Décamps et al. \(2011\)](#) study joint dividend and investment policies. Although these frameworks provide a richer description of corporate behavior, they typically preserve the assumption that default leads directly to liquidation.

Thus, across a wide range of models, whether focused on valuation or on endogenous corporate control, default is commonly treated as an absorbing state. In contrast, other contributions acknowledge that financial distress does not necessarily imply immediate liquidation. Several studies introduce mechanisms in which liquidation is delayed after the firm first enters default. In particular, [Yildırım \(2006\)](#) and [Makarov et al. \(2015\)](#) develop models where liquidation is triggered only after the firm's assets have crossed a threshold and remained below it for a cumulative period of time. From an institutional perspective, [Broadie et al. \(2007\)](#) incorporate bankruptcy procedures into a structural framework, highlighting the economic distinction between reorganization and liquidation. In such settings, default may initiate a restructuring phase rather than an immediate termination of operations.

Despite these advances, the interaction between endogenous corporate policies and a non-instantaneous liquidation mechanism remains only partially explored. Models with delayed liquidation typically focus on valuation problems, whereas stochastic control models of dividend and capital structure decisions usually retain the assumption that default immediately terminates operations.

[Chevalier et al. \(2020\)](#) propose a modeling framework that explicitly distinguishes between default and liquidation. In their setting, the firm holds a fixed level of debt and is subject to a financial covenant. When this covenant is violated, the firm enters technical default but is granted a finite grace period during which equity issuance is allowed in order to restore solvency. If, by the end of this grace period, the firm's assets do not exceed the outstanding debt, liquidation occurs. This framework leads to a mixed stochastic control problem combining **singular controls** (dividend payments and capital injections) with an **optimal stopping decision** (liquidation), and results in a non-standard system of Hamilton–Jacobi–Bellman (HJB) variational inequalities.

The objective of this project is to study and replicate the model introduced by [Chevalier et al. \(2020\)](#). The contributions of this work are threefold. First, we provide a detailed presentation of the modeling framework, including the distinction between financial distress, default, and liquidation, and the associated control constraints. Second, we formulate the dynamic programming principle and the Hamilton–Jacobi–Bellman variational inequalities characterizing the value functions, and discuss their analytical properties, including boundary behavior, monotonicity, and continuity. Third, we implement the finite-difference fixed-point procedure proposed by [Chevalier et al. \(2020\)](#), compare the resulting value functions and intervention regions with the published numerical illustrations, and discuss the sources of any discrepancies.

2 Model setup

This section presents the stochastic control framework introduced by [Chevalier et al. \(2020\)](#). The model describes a firm subject to financial covenants, whose objective is to optimally manage dividend payments and capital injections in the presence of default risk and a finite grace period before liquidation.

2.1 Economic setting

The firm operates with a **fixed level of debt** $D \geq 0$, which is subject to **financial covenants** imposed by its creditors. These covenants are formulated in terms of the firm's asset value and play a key role in determining both the firm's default status and the set of admissible financial decisions.

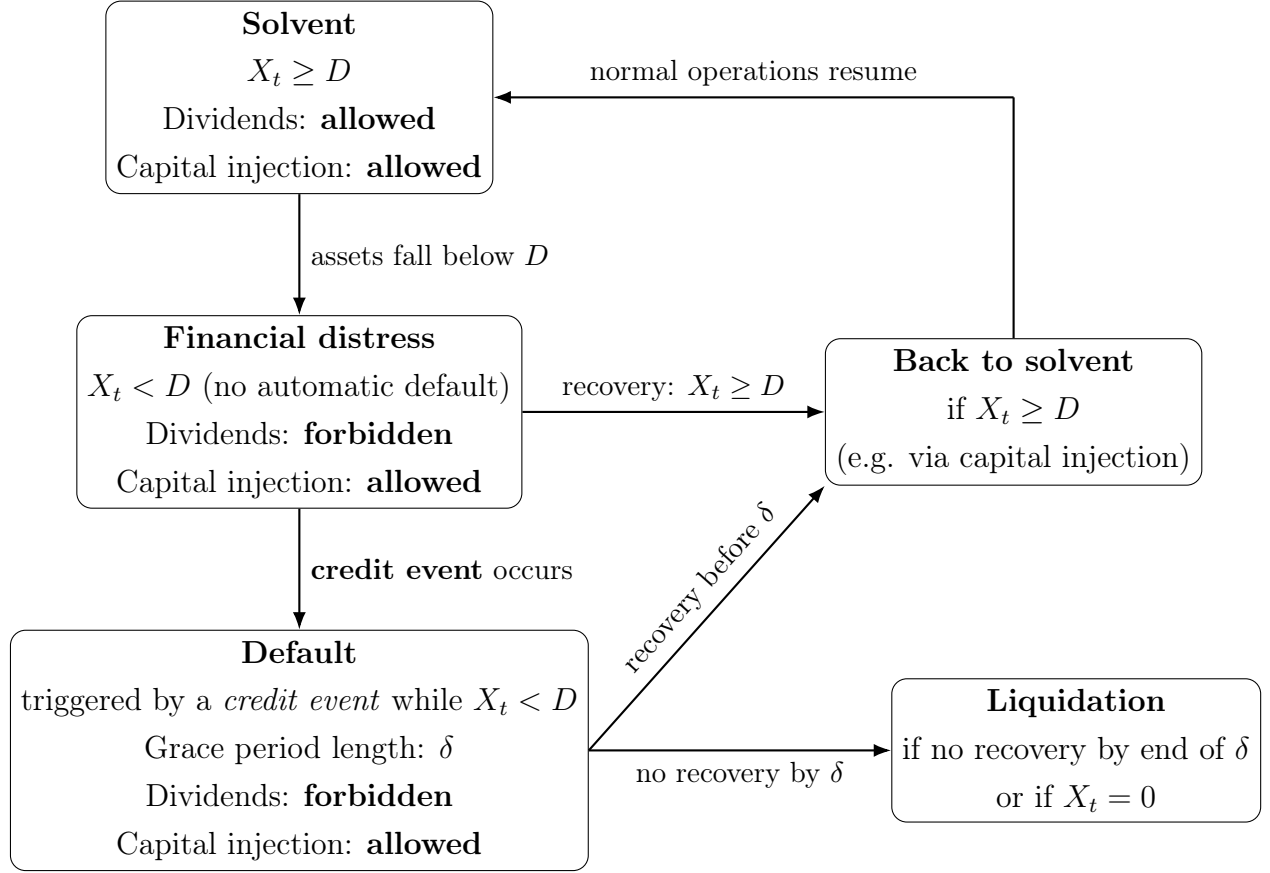
The firm is considered to be in a situation of **financial distress** whenever the value of its assets falls below the outstanding debt. Importantly, financial distress does not automatically lead to default. Instead, **default** is triggered by the occurrence of a **credit event** and may only occur while the firm is already in financial distress. If a credit event occurs when the firm's assets exceed the debt level, the firm remains solvent and does not enter default. As a consequence, the firm may experience periods of financial distress without being formally under default.

Once the firm enters default, it is granted a **grace period** of fixed length $\delta > 0$. During this period, the firm may **take corrective actions** by injecting new capital in order to restore compliance with the financial covenants and recover solvency. If the firm succeeds in exiting default before the end of the grace period, normal operations resume. Otherwise, if the firm fails to recover within the allotted time, or if its asset value reaches zero at any point, the firm is liquidated.

The presence of financial covenants also constrains the firm's control actions. In particular, dividend payments are only allowed when the firm is not in financial distress, whereas capital injections are permitted at all times, including during default. These institutional features are central to the formulation of the firm's optimal dividend and recapitalization policies and motivate the distinction between default and liquidation adopted in this model.

Figure 1 provides a graphical summary of the firm's financial states and the transitions between them under the debt covenant D . In particular, it highlights the distinction between **financial distress** (defined by $X_t < D$) and **default** (which only occurs after a credit event while the firm is already in distress), as well as the role of the grace period δ during which recapitalization may restore solvency. Finally, the diagram emphasizes that dividends are suspended in distress and default, while capital injections remain admissible throughout.

Figure 1: State diagram: financial distress, default with grace period, recovery, and liquidation under debt covenant D . Source: Own elaboration.



2.2 Probability space and sources of uncertainty

All stochastic processes are defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a right-continuous and complete filtration $(\mathcal{F}_t)_{t \geq 0}$, with $\mathcal{F}_0$ containing all $\mathbb{P}$ -null sets. Uncertainty in the model is driven by two **independent** sources. The first is a one-dimensional $(\mathcal{F}_t)$ -**Brownian motion** W , which represents randomness in the **firm's cash flows**. The second is an $(\mathcal{F}_t)$ -adapted **Poisson process** N with constant intensity $\lambda > 0$, which models the occurrence of **credit events**. A jump of the Poisson process corresponds to a credit event and may trigger default when the firm is in financial distress.

2.3 Control processes and asset dynamics

The firm can influence its asset value through two cumulative control processes. The process Z_t denotes the total **amount of dividends** paid up to time t , while K_t represents the **total amount of capital injected** into the firm up to time t . Both processes are assumed to be non-decreasing and $(\mathcal{F}_t)$ -predictable.

The **default status** of the firm is described by a process $I_t \in \{0, 1\}$, where $I_t = 1$ indicates that the firm is under default and $I_t = 0$ otherwise. By construction, default may only be triggered by a credit event when the firm is in financial distress. Dividend payments are suspended during financial distress and default. Accordingly, the dividend control satisfies $dZ_t = 0$ whenever $I_t = 1$.

The evolution of the default status is governed by a jump process driven by credit events. Specifically, the default indicator I_t satisfies:

$$I_t = I_0 + \int_0^t \mathbf{1}_{\{X_s < D\}}(1 - I_{s-}) dN_s - \mathbf{1}_{\{X_t \geq D\}} I_{t-}, \quad (1)$$

where N is a Poisson process with constant intensity $\lambda > 0$. The first term captures *entry into default* following a credit event occurring while the firm is in financial distress, whereas the second term enforces the *immediate exit from default* as soon as the asset value exceeds the debt level D .

The process X_t denotes the firm's asset value and constitutes the main economic state variable of the model. Under the above assumptions, the **controlled asset value process** evolves according to:

$$dX_t = \mu(X_t) dt + \sigma(X_t) dW_t + (1 - \kappa_{I_t}) dK_t - (1 + \kappa') dZ_t, \quad (2)$$

where μ and σ are deterministic functions satisfying standard regularity conditions; $\kappa_0 \leq \kappa_1$ are proportional **capital injection** costs before and during default, respectively; and $\kappa' \in (0, 1)$ is the proportional cost of **dividend payments**.

2.4 Grace period and liquidation time

Let Ξ_t denote the **cumulative amount of time** the firm has spent in default since its last exit from default. Then, when the firm is *not under default*, $\Xi_t = 0$ and when it is *under default*, Ξ_t increases linearly with time and is reset to zero as soon as default is cleared. Hence, Ξ_t records the elapsed time since the beginning of the ongoing default period, ignoring any past default episodes.

The firm is liquidated either if its asset value becomes negative or if the duration of the current default episode exceeds the grace period δ . Accordingly, the **liquidation time** is defined as:

$$T := \inf\{t \geq 0 : X_t < 0 \text{ or } \Xi_t > \delta\}. \quad (3)$$

2.5 State space and admissible strategies

The problem naturally decomposes into two regimes. When the firm is *not under default*, the **state** is fully described by the asset value $x \in S^0 := [0, +\infty)$. When the firm is *under default*, the **state** is described by the pair $(\varsigma, x) \in S^1 := [0, \delta] \times [0, D]$, where ς denotes the elapsed time since the beginning of the current default episode.

The **admissible set of control strategies** is designed to reflect the financial covenants imposed on the firm. In particular, dividend payments are subject to two constraints. First, dividends can only be distributed when the firm's asset value exceeds the outstanding debt, so that no dividends are paid during periods of financial distress. Second, even when the firm is solvent, dividend payments cannot exceed the excess of assets over debt, preventing immediate violation of the solvency constraint. Therefore, the admissible set of dividend distribution and capital injection strategies is defined as:

$$\mathcal{A} = \left\{ (Z, K) \in \mathcal{I}_{\mathbb{F}}^2 : \forall t \geq 0, Z_t - Z_{t-} \leq (X_t - D)^+ \text{ and } \int_0^{+\infty} \mathbf{1}_{\{X_t < D\}} dZ_t = 0 \right\}, \quad (4)$$

where $\mathcal{I}_{\mathbb{F}}$ is the set of non-decreasing, $\mathbb{F}$ -predictable processes of finite variation, and null at time zero.

2.6 Value Functions

The firm's goal is to maximize the expected discounted net cash flow generated by dividend payments minus capital injections, up to the liquidation time T .

No-default regime. When the firm is not under default ($I_t = 0$), the state variable is solely the asset value $x \in S^0 = [0, +\infty)$. In this regime, both dividend payments and capital injections are admissible, subject to the covenant constraints. The corresponding value function is defined as:

$$v_0(x) = \sup_{(Z, K) \in \mathcal{A}} \mathbb{E}_{x,0} \left[\int_0^T e^{-\rho s} d(Z_s - K_s) \right], \quad x \in S^0. \quad (5)$$

Default regime. When the firm is under default ($I_t = 1$), the state becomes two-dimensional. It is described by the pair $(\varsigma, x) \in S^1 = [0, \delta] \times [0, D]$, where x denotes the asset value and ς represents the elapsed time since the beginning of the current default episode (i.e. the remaining grace period is $\delta - \varsigma$). Then, the value function is given by:

$$v_1(\varsigma, x) = \sup_{(Z, K) \in \mathcal{A}} \mathbb{E}_{\varsigma, x, 1} \left[\int_0^T e^{-\rho s} d(Z_s - K_s) \right], \quad (\varsigma, x) \in S^1. \quad (6)$$

3 Dynamic Programming Principle

The optimization problem considered in this project involves **singular controls**, **regime switching** between default and non-default states, and an endogenous **liquidation time**. A natural tool to analyze such problems is the **Dynamic Programming Principle** (DPP), which provides a recursive characterization of the value functions.

Intuitively, the DPP states that, starting from any admissible state, the optimal value of the problem can be decomposed into the expected payoff accumulated up to a given stopping time and the continuation value at that stopping time. This principle applies both before default and during the grace period, although the structure of the continuation value differs across regimes due to the presence of financial covenants and the risk of liquidation.

Let $\tau := \inf\{t \geq 0 : I_t = 1\}$ denote the default time and let T be the liquidation time. Let ν be an arbitrary $\mathcal{F}_t$ -stopping time.

No-default regime. For any initial asset value $x \in S^0$, the value function v_0 satisfies:

$$v_0(x) = \sup_{(Z,K) \in \mathcal{A}} \mathbb{E}_{x,0} \left[\int_0^{\nu \wedge \tau \wedge T} e^{-\rho s} d(Z_s - K_s) + e^{-\rho \nu} v_0(X_\nu) \mathbf{1}_{\{\nu < \tau \wedge T\}} + e^{-\rho \tau} v_1(0, X_\tau) \mathbf{1}_{\{\tau \leq \nu \wedge T\}} \right]. \quad (7)$$

Default regime. For all $(\varsigma, x) \in S^1$ and for $\nu \in [0, \delta - \varsigma]$, the value function v_1 satisfies:

$$v_1(\varsigma, x) = \sup_{(Z,K) \in \mathcal{A}} \mathbb{E}_{\varsigma,x,1} \left[- \int_0^{\nu \wedge \theta^{D,+} \wedge T} e^{-\rho t} dK_t + e^{-\rho \nu} v_1(\Xi_\nu, X_\nu) \mathbf{1}_{\{\nu < T \wedge \theta^{D,+}\}} + e^{-\rho \theta^{D,+}} v_0(X_{\theta^{D,+}}) \mathbf{1}_{\{\theta^{D,+} \leq \nu \wedge T\}} \right]. \quad (8)$$

where $\theta^{D,+} = \inf\{t \geq 0 : X_t \geq D\}$.

The value functions inherit several structural properties from the control problem and the DPP. First, both v_0 and v_1 are **non-negative** and **non-decreasing** with respect to the asset value, reflecting the fact that additional assets cannot reduce the firm's continuation value. Moreover, for any fixed asset level $x \in [0, D]$, the post-default value function $v_1(\varsigma, x)$ is non-increasing in the remaining grace period ς , since a shorter time to liquidation reduces the opportunity to recover solvency.

Second, the value functions satisfy **natural boundary conditions**. At the end of the grace period, the post-default value function coincides with the terminal payoff,

$$v_1(\delta, x) = g(x), \quad x \in [0, D].$$

At the solvency boundary $x = D$, the value functions match continuously,

$$v_1(\varsigma, D) = v_0(D), \quad \varsigma \in [0, \delta],$$

ensuring consistency between regimes.

Finally, under standard regularity assumptions on the drift and volatility coefficients, both value functions are **continuous** on their respective domains, namely $v_0 \in C(S^0)$ and $v_1 \in C(S^1)$. These properties guarantee that the HJB variational inequalities introduced in the next section are well posed and admit a viscosity solution characterization.

3.1 Hamilton-Jacobi-Bellman equation

Let $\mathcal{L}$ denote the infinitesimal generator associated with the uncontrolled asset value process. For any twice continuously differentiable function φ , the operator $\mathcal{L}$ is given by:

$$\mathcal{L}\varphi(x) = \mu(x)\varphi'(x) + \frac{1}{2}\sigma^2(x)\varphi''(x). \quad (9)$$

This operator describes the local evolution of the asset value in the absence of dividend payments and capital injections and appears naturally in the HJB equations through the application of Itô's formula. The two regimes are coupled through the jump operator:

$$\mathcal{J}(v_0, v_1)(x) = \lambda \mathbf{1}_{[0, D]}(x) (v_1(0, x) - v_0(x)), \quad x \in S^0, \quad (10)$$

which accounts for the possibility of a credit event triggering default when the firm is in financial distress. The evaluation of v_1 at elapsed time $\varsigma = 0$ reflects the fact that any new default episode begins at elapsed time zero.

Before Default. When the firm is not under default, the value function v_0 satisfies a variational inequality reflecting three possible actions: continuation of operations, payment of dividends, and injection of capital. The corresponding HJB equation takes the form:

$$\min \left\{ \rho v_0(x) - \mathcal{L}v_0(x) - \mathcal{J}(v_0, v_1)(x), \quad v_0'(x) - \frac{1}{1 + \kappa'}, \quad \frac{1}{1 - \kappa_0} - v_0'(x) \right\} = 0, \quad x \in S^0. \quad (11)$$

The first term corresponds to the continuation region, where it is optimal to neither pay dividends nor inject capital; it includes the jump term $\mathcal{J}(v_0, v_1)$ which captures the expected

loss in value upon a credit event while the firm is in financial distress. The second term characterizes the dividend region, in which the marginal value of assets equals the marginal benefit of dividend payments. The third term corresponds to the capital injection region, where injecting capital becomes optimal.

During Default. During the grace period, the value function v_1 depends on both the asset value x and the elapsed time ς since the beginning of the current default episode. In this regime, the HJB equation includes an additional time derivative and reflects the possibility of recapitalization or liquidation. The corresponding variational inequality is given by:

$$\min \left\{ \rho v_1(\varsigma, x) - \mathcal{L}v_1(\varsigma, x) - \partial_\varsigma v_1(\varsigma, x), \frac{1}{1 - \kappa_1} - \partial_x v_1(\varsigma, x) \right\} = 0, \quad (\varsigma, x) \in S^1. \quad (12)$$

The first term corresponds to continuation under default, while the second term characterizes the region in which capital injection is optimal. Dividend payments are not allowed during default and therefore do not appear in the post-default HJB equation. Note that the jump operator $\mathcal{J}$ does not appear here because, once in default, the firm cannot re-enter default from the same episode; the only transitions are recovery (exit to S^0) or liquidation.

4 Characterization of optimal policies

In this section, we characterize the optimal dividend and capital injection strategies implied by the variational inequalities derived in section 3. The structure of the problem differs substantially across the pre-default and default regimes. We therefore analyze these two cases separately.

4.1 Optimal strategies during default

When the firm is in default, dividend payments are no longer allowed. In this case, the value function $v_1(\varsigma, x)$ satisfies the variational inequality (12), and only two regions arise:

- The **continuation region under default**, where the firm waits for a possible recovery of asset value:

$$\mathcal{C}_1 = \{(\varsigma, x) \in S^1 : \rho v_1(\varsigma, x) - \mathcal{L}v_1(\varsigma, x) - \partial_\varsigma v_1(\varsigma, x) = 0\}$$

- The **recapitalization region**, where capital injection is optimal in order to restore solvency and exit default:

$$\mathcal{K}_1 = \left\{ (\varsigma, x) \in S^1 : \partial_x v_1(\varsigma, x) = \frac{1}{1 - \kappa_1} \right\}.$$

Since dividend payments are not allowed during default, the **only admissible control** is **capital injection**. The goal is to restore solvency before the end of the grace period δ .

Let D denote the **debt level** and let $\kappa_1 \in (0, 1)$ be the proportional **issuance cost** in default. The following auxiliary functions on $\mathbb{R}^+$ are introduced by [Chevalier et al. \(2020\)](#):

$$g(x) = \left(v_0(D) - \frac{D-x}{1-\kappa_1} \right)^+, \quad h(x) = \frac{1}{1-\kappa_1} \left(x + \frac{\mu}{\rho} \right), \quad (13)$$

where the function g corresponds to the payoff obtained when injecting just enough capital to bring assets up to D , while h is a convenient explicit function that will serve as a supersolution and an upper bound for the default value function.

Following [Chevalier et al. \(2020\)](#), we also define the (time-independent) threshold:

$$x_1^* := (1 - \kappa_1)(h(D) - g(D)) = D + \frac{\mu}{\rho} - (1 - \kappa_1)v_0(D). \quad (14)$$

Case 1: $x_1^* \leq 0$. Immediate recapitalization.

A first important result is that if $x_1^* \leq 0$, then *capital injection is optimal everywhere in default*:

$$v_1 = g \quad \text{on } S^1, \quad \mathcal{K}^1 = S^1. \quad (15)$$

Therefore, the condition $x_1^* \leq 0$ is equivalent to:

$$(1 - \kappa_1)v_0(D) \geq D + \frac{\mu}{\rho}. \quad (16)$$

which means that the value of restoring solvency, net of issuance costs, dominates the continuation value in default. In other words, the continuation value in default is never high enough to justify waiting. As a consequence, it is always optimal to inject capital immediately, independently of the current asset level. The firm restores its assets to the debt level D as soon as default occurs.

Case 2: $x_1^* > 0$. Interior recapitalization threshold.

Now, we focus on the economically relevant case $x_1^* > 0$, which is equivalent to:

$$v_0(D) < h(D) = \frac{1}{1-\kappa_1} \left(D + \frac{\mu}{\rho} \right). \quad (17)$$

In this regime, **recapitalization is not always optimal**. Instead, the optimal strategy is characterized by the time-dependent boundary.

Let $\varsigma \in [0, \delta]$ denote the remaining time until the end of the grace period. The optimal policy in default is characterized by a *time-dependent free boundary* $x_1^*(\varsigma)$ defined as the smallest solution on $[0, D]$ of:

$$v_1(\varsigma, x) = g(x) = \left(v_0(D) - \frac{D - x}{1 - \kappa_1} \right)^+. \quad (18)$$

The boundary $x_1^*(\varsigma)$ is *non-increasing* in ς , meaning that recapitalization becomes more likely as the end of the grace period approaches. This boundary separates the default continuation region from the capital injection region. Moreover, the capital issuance region in default has the following structure:

$$\mathcal{K}^1 = [0, \varsigma^*) \times \{0\} \cup \{(\varsigma, x) \in S^1 : x \geq x_1^*(\varsigma)\}. \quad (19)$$

The default region is divided into three areas:

- **Exit-default recapitalization.** If $x \geq x_1^*(\varsigma)$, it is optimal to inject capital so as to restore the asset level to D (net of issuance costs) and exit default. In this region, the benefit of restoring solvency exceeds the continuation value.

$$\mathcal{K}_1^1 := \{(\varsigma, x) \in S^1 : x \geq x_1^*(\varsigma)\}. \quad (20)$$

- **Continuation region.** If $0 < x < x_1^*(\varsigma)$, the firm waits and does not inject capital. In this region, the expected continuation value dominates the immediate gain from recapitalization.
- **Survival at zero.** If the firm hits $x = 0$ with sufficient remaining time, it is optimal to inject minimally to keep X just above zero and avoid immediate bankruptcy, hoping that the diffusion evolves favorably and reaches the exit-default region $\mathcal{K}_1^1$. For $\varsigma \geq \varsigma^*$, there is insufficient time to recover and liquidation becomes optimal.

$$\mathcal{K}_2^1 := [0, \varsigma^*) \times \{0\}. \quad (21)$$

Overall, recapitalization is optimal either (i) when assets are close enough to D to justify injecting up to solvency, or (ii) at the boundary $x = 0$ when enough time remains to potentially recover before the grace period ends.

Finally, the optimal policy is of threshold type and can be written as:

$$\pi^*(\varsigma, x) = \begin{cases} \text{inject capital up to } D \text{ (exit default),} & \text{if } x \geq x_1^*(\varsigma), \\ \text{wait (no injection),} & \text{if } 0 < x < x_1^*(\varsigma), \\ \text{minimal injection to keep } X \gtrsim 0, & \text{if } x = 0 \text{ and } \varsigma < \varsigma^*, \\ \text{liquidate (no injection),} & \text{if } x = 0 \text{ and } \varsigma \geq \varsigma^*. \end{cases} \quad (22)$$

4.2 Optimal policies before default

When the firm is not under default, the value function v_0 satisfies the variational inequality (11). Then three regions arise:

- The **continuation region**, where it is optimal to neither distribute dividends nor inject capital:

$$\mathcal{C}_0 = \{x \in S^0 : \rho v_0(x) - \mathcal{L}v_0(x) = 0\} \quad (23)$$

- The **dividend region**, where the marginal value of assets equals the effective marginal benefit of dividend payments:

$$\mathcal{D} = \left\{ x \in S^0 : v'_0(x) = \frac{1}{1 + \kappa'} \right\} \quad (24)$$

- The **capital injection region**, where issuing equity becomes optimal:

$$\mathcal{K}_0 = \left\{ x \in S^0 : v'_0(x) = \frac{1}{1 - \kappa_0} \right\} \quad (25)$$

Here, the optimal strategy is characterized by two policies: a dividend policy and a capital injection policy.

Dividend policy. There exists a threshold $x_0^d \geq D$ such that the dividend region is given by $\mathcal{D} = [x_0^d, +\infty)$. More precisely, the optimal dividend threshold x_0^d is given by:

$$x_0^d = \begin{cases} D, & \text{if } v_0(D) \geq \frac{\mu}{(1 + \kappa')\rho}, \\ \text{the unique solution in } (D, +\infty) \text{ of } \rho v_0(x) = \frac{\mu}{1 + \kappa'}, & \text{if } v_0(D) < \frac{\mu}{(1 + \kappa')\rho}. \end{cases}$$

On (D, ∞) , the function v_0 is *concave* and *twice continuously differentiable*, and becomes *linear* in the dividend region. Economically, when the firm's assets exceed x_0^d , it is optimal to distribute all excess cash as dividends.

Capital injection policy. Capital issuance is never optimal when the firm's asset value strictly exceeds the debt level D . Formally,

$$\mathcal{K}^0 \cap (D, +\infty) = \emptyset. \quad (26)$$

Hence, in the solvent region above D , the firm either continues or distributes dividends, but never raises additional capital.

Below the debt level D , the structure of the capital injection region depends on the parameter x_1^* , given by expression (14), which determines whether recapitalization is immediate or occurs only in a subinterval of $(0, D)$.

Case 1: $x_1^* \leq 0$. Immediate recapitalization.

In this case, the optimal policy prescribes immediate capital injection whenever the firm's asset value falls below the debt level D . The capital issuance region coincides with the entire interval $(0, D)$:

$$\mathcal{K}^0 = (0, D). \quad (27)$$

The value function on $[0, D]$ is given by:

$$v_0(x) = \left(v_0(D) - \frac{D-x}{1-\kappa_0} \right)^+, \quad x \in [0, D]. \quad (28)$$

Economically, the firm restores its asset value to the debt level D as soon as it falls below it, thereby avoiding the risk of entering default under less favorable conditions.

Then, the optimal policy can be summarized as follows:

$$\pi^*(x) = \begin{cases} \text{capital injection,} & \text{if } 0 < x < D, \\ \text{continuation,} & \text{if } D \leq x < x_0^d, \\ \text{dividend payments,} & \text{if } x \geq x_0^d. \end{cases} \quad (29)$$

Case 2: $x_1^* > 0$. Interior recapitalization threshold.

In this case, recapitalization occurs only within a subinterval of $[0, D]$. There exists a threshold $x_0^* \in [0, D]$, defined by:

$$x_0^* = \sup \left\{ x \in [0, D] : v_0'(x) < \frac{1}{1-\kappa_0} \right\}, \quad (30)$$

such that $(x_0^*, D) \in \mathcal{K}^0$. Equivalently, x_0^* is the smallest solution of:

$$v_0(x) = \left(v_0(D) - \frac{D - x}{1 - \kappa_0} \right)^+. \quad (31)$$

Then, the optimal policy can be summarized as follows:

$$\pi^*(x) = \begin{cases} \text{capital injection,} & \text{if } 0 < x \leq x_0^*, \\ \text{continuation,} & \text{if } x_0^* < x < x_0^d, \\ \text{dividend payments,} & \text{if } x \geq x_0^d. \end{cases} \quad (32)$$

In this regime, capital is injected only when the asset value is sufficiently close to the debt level D . For lower values of x , it is optimal to wait rather than recapitalize immediately.

Hence, before default, the firm's optimal policy follows a threshold structure characterized by two free boundaries x_0^* and x_0^d . The firm behaves as in a classical singular control problem with asymmetric transaction costs and covenant restrictions.

4.3 Comparison of regimes

The characterization obtained in the previous subsections shows a fundamental structural difference between the two regimes.

Table 1 highlights the main differences between the two regimes. **Before default**, the problem is stationary and the optimal policy follows a double-threshold structure. Capital is injected when assets are too low, dividends are distributed when assets are sufficiently high, and no action is taken in between. The optimal decision depends only on the current asset level. **During default**, the structure changes because of the finite grace period. Dividend payments are suspended and the firm can only decide whether or not to recapitalize. The optimal recapitalization threshold becomes time-dependent and varies with the remaining duration of the grace period. As the end of the grace period approaches, waiting becomes less attractive and recapitalization becomes more likely. This time dependence is the key structural difference between the default and pre-default regimes.

Table 1: Comparison of optimal policies before and during default. Source: own elaboration.

During Default (v_1)	Before Default (v_0)
$(\varsigma, x) \in S^1$ (remaining grace period and assets).	$x \in S^0$ (assets only).
Time-dependent free boundary $x_1^*(\varsigma)$ (non-increasing in ς).	Stationary thresholds x_0^* and x_0^d .
Admissible control: only capital injection.	Admissible controls: dividend payments and capital injection.
Dividend region not allowed.	$\mathcal{D} = [x_0^d, +\infty)$.
Capital injection region: $x \geq x_1^*(\varsigma)$ and possibly $x = 0$ if $\varsigma < \varsigma^*$.	Capital injection region: $0 < x \leq x_0^*$ if $x_1^* > 0$.
Continuation region: $0 < x < x_1^*(\varsigma)$.	Continuation region: $x_0^* < x < x_0^d$.
$\min\{\rho v_1 - \mathcal{L}v_1 - \partial_\varsigma v_1, \partial_x v_1 - \frac{1}{1-\kappa_1}\} = 0$.	$\min\{\rho v_0 - \mathcal{L}v_0 - \mathcal{J}(v_0, v_1), v_0' - \frac{1}{1+\kappa'}, \frac{1}{1-\kappa_0} - v_0'\} = 0$.
Dynamic trade-off between waiting and recapitalizing before the end of the grace period.	Double-barrier stationary policy: inject capital below x_0^* , distribute dividends above x_0^d .
If $x_1^* \leq 0$: recapitalization everywhere.	If $x_1^* \leq 0$: capital injection on $(0, D)$.

5 Methodology: Preliminary Pre-Default Solver

In this section, we present a first numerical implementation restricted to the **pre-default regime only**, treating the boundary value $v_0(D)$ as a fixed external parameter. This serves as a validation step for the policy iteration scheme before introducing the full coupled solver in Section 6.

5.1 Model specification

We assume constant drift and volatility coefficients: $\mu(x) = \mu$ and $\sigma(x) = \sigma$. Under this specification, the infinitesimal generator becomes:

$$\mathcal{L}v(x) = \mu v'(x) + \frac{\sigma^2}{2} v''(x). \quad (33)$$

The pre-default variational inequality to be solved is:

$$\min \left\{ \rho v_0(x) - \mathcal{L}v_0(x), v'_0(x) - \frac{1}{1 + \kappa'}, \frac{1}{1 - \kappa_0} - v'_0(x) \right\} = 0, \quad x \in S^0. \quad (34)$$

The state space $S^0 = [0, +\infty)$ is truncated to $[0, X_{\max}]$. The numerical parameters used in this section are reported in Table 2.

Table 2: Parameter values for the standalone pre-default solver of Section 5.

Parameter	Symbol	Value
Discount rate	ρ	0.05
Drift	μ	0.10
Volatility	σ	0.20
Capital injection cost	κ_0	0.05
Dividend cost	κ'	0.02
Domain truncation	$X_{\max}$	10.0
Grid points	N	800

5.2 Spatial discretization

We introduce a uniform spatial grid on $[0, X_{\max}]$ defined by

$$x_i = i\Delta x, \quad i = 0, \dots, N, \quad \Delta x = \frac{X_{\max}}{N}. \quad (35)$$

The operator $\mathcal{L}$ is approximated by a central finite difference scheme on interior points:

$$\mathcal{L}v(x_i) \approx \mu \frac{v_{i+1} - v_{i-1}}{2\Delta x} + \frac{\sigma^2}{2} \frac{v_{i+1} - 2v_i + v_{i-1}}{\Delta x^2}. \quad (36)$$

Remark 5.1. This section uses a central difference for the drift term, which is adequate for the standalone pre-default problem where no CFL stability constraint is active. The full coupled solver of Section 6 replaces this with a monotone upwind scheme to ensure convergence toward the viscosity solution of the coupled system.

Boundary conditions are imposed as follows: $v_0(0) = 0$ and $v'_0(X_{\max}) = 1/(1 + \kappa')$, consistent with the dividend region at large asset levels.

5.3 Resolution of the variational inequality

At each grid point x_i , the discretized problem becomes:

$$\min\{\mathcal{C}_i(v), \mathcal{D}_i(v), \mathcal{K}_i(v)\} = 0, \quad (37)$$

where $\mathcal{C}_i(v)$ is the discretized continuation equation, $\mathcal{D}_i(v)$ is the dividend constraint, and $\mathcal{K}_i(v)$ is the capital injection constraint. For the gradient constraints, one-sided differences are used to preserve monotonicity: backward differences in the dividend region and forward differences in the capital injection region.

The numerical resolution proceeds by policy iteration:

1. Start from an initial guess $v^{(0)}(x) = x/(1 + \kappa')$.
2. At each grid point, identify the active constraint by evaluating all three operators and selecting the one closest to zero.
3. Solve the resulting linear system.
4. Update the policy and repeat until both the value function and the active policy stabilize.

5.4 Identification of optimal thresholds

Once convergence is achieved, the optimal regions are identified from the discrete derivative of the value function:

$$x_K^* = \max\{x_i : v'_0(x_i) \approx \frac{1}{1 - \kappa_0}\}, \quad x_D^* = \min\{x_i : v'_0(x_i) \approx \frac{1}{1 + \kappa'}\}. \quad (38)$$

The results of this standalone solver, including the value function, derivative plot, optimal regions, and thresholds, are reported in Section 7 together with the full coupled results. The full coupled solver, which jointly resolves both regimes through a fixed-point bisection on $a^* = v_0(D)$, is presented in Section 6.

6 Methodology: Coupled Fixed-Point Solver

Building on the standalone pre-default scheme of Section 5, we now present the full numerical procedure that jointly resolves the default and pre-default regimes. The value functions (v_0, v_1) satisfy, in the viscosity sense, the coupled system of variational inequalities. Due to the presence of singular controls, regime switching, and a finite grace period, the problem combines a stationary singular control problem (pre-default) with a parabolic obstacle problem (default regime). The main difficulty is that the two domains S^0 and S^1 are coupled through the unknown boundary value $a := v_0(D)$. Following [Chevalier et al. \(2020\)](#), we reformulate the system as a fixed-point problem in a , solved by bisection.

6.1 Discretization

Since $S^0 = [0, +\infty)$ is unbounded, we truncate the spatial domain to $[0, x_{\max}]$, where $x_{\max} \gg D$ is chosen sufficiently large. We introduce a uniform spatial grid defined by:

$$x_i = i\Delta x, \quad i = 0, \dots, N_x, \quad \Delta x = \frac{x_{\max}}{N_x}. \quad (39)$$

Let i_D denote the index such that $x_{i_D} = D$ (or the closest grid point).

On the default region $S^1 = [0, \delta] \times [0, D]$, we use the same spatial discretization on $[0, D]$ and introduce a uniform grid in elapsed default time:

$$\varsigma_n = n\Delta\varsigma, \quad n = 0, \dots, N_\varsigma, \quad \Delta\varsigma = \frac{\delta}{N_\varsigma}. \quad (40)$$

Under constant coefficients, the infinitesimal generator is:

$$\mathcal{L}v = \mu \partial_x v + \frac{1}{2} \sigma^2 \partial_{xx} v. \quad (41)$$

To ensure monotonicity of the numerical scheme and guarantee convergence toward the viscosity solution, we discretize the first-order term using a **upwind** (one-sided) finite difference whose direction depends on the sign of the drift μ , while the second-order term uses a standard centered difference:

$$(D_x^h V)_i = \begin{cases} \frac{V_i - V_{i-1}}{\Delta x}, & \text{if } \mu \geq 0, \\ \frac{V_{i+1} - V_i}{\Delta x}, & \text{if } \mu < 0, \end{cases} \quad (D_{xx}^h V)_i = \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta x^2}. \quad (42)$$

Defining $\mu^+ = \max(\mu, 0)$ and $\mu^- = \max(-\mu, 0)$, the discrete diffusion operator takes the form:

$$(\mathcal{L}_h V)_i = a_\ell V_{i-1} + a_m V_i + a_u V_{i+1}, \quad (43)$$

where the coefficients are given by:

$$a_\ell = \frac{\sigma^2}{2\Delta x^2} + \frac{\mu^-}{\Delta x}, \quad a_m = -\frac{\sigma^2}{\Delta x^2} - \frac{\mu^+ + \mu^-}{\Delta x}, \quad a_u = \frac{\sigma^2}{2\Delta x^2} + \frac{\mu^+}{\Delta x}. \quad (44)$$

Note that $a_\ell, a_u \geq 0$ and $a_m \leq 0$ for all $\Delta x > 0$, which ensures the monotonicity of the scheme. This upwind discretization is applied uniformly in both the default solver (backward sweep in ς) and the pre-default solver (stationary policy iteration).

6.2 Default Regime

First, we fixed parameter $a > 0$ and we denote by v_1^a the solution of the following variational inequality on $S^1 = [0, \delta] \times [0, D]$. In the interior of the domain $\mathring{S}^1 = (0, \delta) \times (0, D)$, the value function satisfies:

$$\min(\rho v - \mathcal{L}v - \partial_\varsigma v, g^a(x) - v) = 0, \quad (\varsigma, x) \in \mathring{S}^1, \quad (45)$$

where the obstacle function is defined by:

$$g^a(x) := \left(a - \frac{D - x}{1 - \kappa_1}\right)^+, \quad x \in [0, D]. \quad (46)$$

The boundary condition at the end of the grace period is given by:

$$v(\delta, x) = g^a(x), \quad x \in [0, D]. \quad (47)$$

At the boundary $x = 0$, the value function satisfies the gradient: constraint:

$$\min\left(v, \frac{1}{1 - \kappa_1} - \partial_x v\right) = 0, \quad \text{on } [0, \delta) \times \{0\}. \quad (48)$$

In order to solve this parabolic obstacle problem numerically, we employ an explicit *finite-difference scheme* in the time variable ς and a monotone discretization of the operator $\mathcal{L}$.

Let (ς_n, x_i) denote the discrete grid with step sizes $\Delta\varsigma$ and Δx . Starting from the terminal condition at $\varsigma = \delta$, we proceed backward in ς . At each time level ς_n , the numerical procedure consists of two projection steps:

1. **Obstacle projection.** After computing the explicit update of the linear equation:

$$v_{n,i}^{\text{PDE}} = v_{n+1,i} - \Delta\varsigma(\rho v_{n+1,i} - \mathcal{L}_h v_{n+1,i}), \quad (49)$$

we enforce the obstacle condition by setting:

$$v_{n,i} = \max(v_{n,i}^{\text{PDE}}, g^a(x_i)). \quad (50)$$

2. **Boundary projection at $x = 0$.** The variational inequality (45) implies that, at $x = 0$, either $v_1^a(\varsigma, 0) = 0$ (no capital injection) or:

$$\partial_x v_1^a(\varsigma, 0) = \frac{1}{1 - \kappa_1} \quad (\text{marginal capital injection}). \quad (51)$$

Approximating the spatial derivative by a *forward difference*:

$$\partial_x v_1^a(\varsigma, 0) \approx \frac{v_{n,1} - v_{n,0}}{\Delta x}. \quad (52)$$

Therefore, at each time step we compare the two admissible boundary values and retain the largest:

$$v_{n,0} = \max\left(0, v_{n,1} - \frac{\Delta x}{1 - \kappa_1}\right). \quad (53)$$

This projection ensures that the discrete solution satisfies the gradient constraint at $x = 0$ while preserving monotonicity of the scheme. The family $\{v_1^a\}_{a>0}$ is increasing with respect to a . If $a \leq b$, then $v_1^a \leq v_1^b$ on S^1 , since the obstacle g^a is increasing in a and the variational inequality is monotone. Moreover, the solutions satisfy the Lipschitz estimate:

$$0 \leq v_1^b - v_1^a \leq b - a, \quad \text{on } S^1. \quad (54)$$

This property guarantees that the map $a \mapsto v_0^a(D)$ is Lipschitz and non-decreasing, which ensures existence and uniqueness of the fixed point.

6.3 Pre-default

Pre-default on $[0, D)$. Once the function v_1^a has been computed, we determine the function v_0^a as the solution of the following variational inequality on $[0, D)$:

$$\min\left(\rho v - \mathcal{L}v - \mathcal{J}(v, v_1^a), \frac{1}{1 - \kappa_0} - v'\right) = 0, \quad x \in (0, D), \quad (55)$$

where the coupling operator is defined by:

$$\mathcal{J}(v, v_1^a)(x) = \lambda(v_1^a(0, x) - v(x)). \quad (56)$$

The boundary conditions are given by:

$$\min \left(v(0), \frac{1}{1 - \kappa_0} - v'(0) \right) = 0, \quad v(D) = a. \quad (57)$$

The problem is solved on a spatial grid:

$$x_i = i\Delta x, \quad i = 0, \dots, N, \quad x_N = D. \quad (58)$$

Since the equation is stationary, we solve directly the discretized variational inequality using a monotone finite-difference approximation $\mathcal{L}_h$ of the operator $\mathcal{L}$.

1. **Gradient projection.** We first compute the discrete solution of the linear equation:

$$\rho v - \mathcal{L}v - \mathcal{J}(v, v_1^a) = 0, \quad (59)$$

which yields an intermediate value v_i^{PDE} . Then, the singular constraint imposes:

$$v'(x) \geq \frac{1}{1 - \kappa_0}. \quad (60)$$

Approximating the derivative by a *backward finite difference*:

$$v'(x_i) \approx \frac{v_i - v_{i-1}}{\Delta x}, \quad (61)$$

the constraint becomes:

$$\frac{v_i - v_{i-1}}{\Delta x} \geq \frac{1}{1 - \kappa_0}. \quad (62)$$

Therefore, we project onto the admissible set:

$$v_i = \max \left(v_i^{\text{PDE}}, v_{i-1} + \frac{\Delta x}{1 - \kappa_0} \right). \quad (63)$$

This ensures that the discrete derivative satisfies the required slope constraint while preserving monotonicity of the scheme.

2. **Boundary projection at $x = 0$.** The variational inequality (55) implies that, at $x = 0$, either $v(0) = 0$ or:

$$v'(0) = \frac{1}{1 - \kappa_0}. \quad (64)$$

Approximating the derivative by a *forward difference*:

$$v'(0) \approx \frac{v_1 - v_0}{\Delta x}, \quad (65)$$

the slope condition yields:

$$v_0 = v_1 - \frac{\Delta x}{1 - \kappa_0}. \quad (66)$$

Therefore, at each iteration we compare the two admissible values and retain the largest:

$$v_0 = \max \left(0, v_1 - \frac{\Delta x}{1 - \kappa_0} \right). \quad (67)$$

This projection guarantees consistency with the variational inequality while preserving monotonicity of the numerical scheme. As in the default regime, the family $\{v_0^a\}_{a \geq 0}$ is monotone with respect to the boundary datum $a = v_0(D)$. More precisely, for any $a \leq b$, $v_0^a \leq v_0^b$ on $[0, D]$. Moreover, using the Lipschitz estimate (54) for v_1^a , one can show that, for $b \geq a$, the function $v_0^a + (b - a)$ is a supersolution of the HJB equation satisfied by v_0^b . By the comparison principle, this yields the Lipschitz bound:

$$0 \leq v_0^b(x) - v_0^a(x) \leq b - a, \quad x \in [0, D], \quad \text{for any } a \leq b. \quad (68)$$

Pre-default on $(D, +\infty)$. We now compute v_0^a on $(D, +\infty)$ as the solution of the stationary variational inequality:

$$\min \left(\rho v - \mathcal{L}v, v' - \frac{1}{1 + \kappa'}, \frac{1}{1 - \kappa_0} - v' \right) = 0, \quad x \in (D, +\infty), \quad (69)$$

with boundary condition:

$$v(D) = a. \quad (70)$$

The problem is solved on the spatial grid:

$$x_i = i\Delta x, \quad i = i_D, \dots, N_x, \quad x_{i_D} = D. \quad (71)$$

As in $[0, D)$, we use a monotone finite-difference approximation $\mathcal{L}_h$ of the operator $\mathcal{L}$.

1. Gradient projection.

We first compute the discrete solution of the linear equation

$$\rho v - \mathcal{L}v = 0, \quad (72)$$

which yields an intermediate value v_i^{PDE} . The variational inequality imposes the slope constraints:

$$\frac{1}{1 + \kappa'} \leq v'(x) \leq \frac{1}{1 - \kappa_0}. \quad (73)$$

Note that $1/(1 + \kappa') < 1/(1 - \kappa_0)$ for any $\kappa_0, \kappa' > 0$, so this interval is non-empty. Economically, the lower bound corresponds to the dividend slope and the upper bound to the capital injection slope; by Lemma 4.5 of [Chevalier et al. \(2020\)](#), injection is never optimal above D , so in practice only the lower bound is active.

Approximating the derivative by a *backward finite difference*:

$$v'(x_i) \approx \frac{v_i - v_{i-1}}{\Delta x}, \quad (74)$$

the discrete constraint becomes:

$$\frac{\Delta x}{1 + \kappa'} \leq v_i - v_{i-1} \leq \frac{\Delta x}{1 - \kappa_0}. \quad (75)$$

Therefore, we project onto the admissible interval:

$$v_i = \min \left(\max \left(v_i^{\text{PDE}}, v_{i-1} + \frac{\Delta x}{1 + \kappa'} \right), v_{i-1} + \frac{\Delta x}{1 - \kappa_0} \right). \quad (76)$$

2. Boundary condition at $x = D$.

At the coupling point we impose directly:

$$v_{i_D} = a. \quad (77)$$

3. Right boundary at $x = x_{\max}$.

At the truncation point $x_{N_x} = x_{\max}$, the dividend region is characterized by the slope constraint:

$$v'(x_{\max}) = \frac{1}{1 + \kappa'}. \quad (78)$$

Using a backward difference, this yields:

$$v_{N_x} = v_{N_x-1} + \frac{\Delta x}{1 + \kappa'}. \quad (79)$$

We denote again by v_0^a the solution on $(D, +\infty)$, so that v_0^a is defined on the whole pre-default state space $S^0 = [0, +\infty)$. As before, monotonicity and stability arguments imply that for any $a \leq b$,

$$0 \leq v_0^b(x) - v_0^a(x) \leq b - a, \quad x \in S^0. \quad (80)$$

6.4 Fixed point in a : existence and uniqueness

The pre-default and default regimes are coupled through the value of the function at the default threshold D . More precisely, for a given parameter $a \geq 0$, we solve the default problem and then the pre-default variational inequality imposing the boundary condition:

$$v(D) = a. \quad (81)$$

However, the parameter a must be consistent with the solution itself. Therefore, the correct value is characterized by the fixed-point condition:

$$v_0^a(D) = a. \quad (82)$$

To formalize this, we introduce the scalar function:

$$F(a) := v_0^a(D) - a. \quad (83)$$

The full coupled HJB system is thus reduced to finding a zero of F .

From the global Lipschitz estimate (54), we have:

$$0 \leq v_0^b(x) - v_0^a(x) \leq b - a, \quad x \in S^0, \quad a \leq b. \quad (84)$$

In particular,

$$0 \leq v_0^b(D) - v_0^a(D) \leq b - a. \quad (85)$$

Hence the mapping $a \mapsto v_0^a(D)$ is 1-Lipschitz and non-decreasing, which implies that F is continuous and non-increasing. Moreover, $F(0) = v_0^0(D) \geq 0$.

In order to study the sign of $F(a)$ for large values of a , let:

$$a \geq \max\left(\frac{\mu}{\kappa'\rho}, D\right) \quad (86)$$

and define on S^0 the function:

$$\psi^a(x) := \frac{a}{\mu + \rho a} (\rho x + \mu). \quad (87)$$

For a sufficiently large, one checks that the slope belongs to the admissible interval:

$$\frac{1}{1 + \kappa'} \leq (\psi^a)'(x) = \frac{\rho a}{\mu + \rho a} \leq \frac{1}{1 - \kappa_0}. \quad (88)$$

As a consequence, ψ^a is a supersolution of the HJB variational inequalities satisfied by v_0^a , and by comparison we obtain the upper bound $v_0^a(x) \leq \psi^a(x)$, $x \in S^0$.

In particular, evaluating at $x = D$ gives:

$$F(a) = v_0^a(D) - a \leq \psi^a(D) - a = -\frac{\rho a}{\mu + \rho a}(a - D) \leq 0. \quad (89)$$

Therefore, since $F(0) \geq 0$, F is continuous and non-increasing, and $F(a) \leq 0$ for a large

enough, there exists a unique fixed point a^* such that:

$$F(a^*) = 0, \quad \text{i.e.} \quad v_0^{a^*}(D) = a^*. \quad (90)$$

Moreover, any solution must belong to the interval:

$$\left[0, \max\left(\frac{\mu}{\kappa'\rho}, D\right)\right]. \quad (91)$$

Finally, uniqueness of a^* is consistent with the comparison result for the HJB system, which guarantees that the coupled variational inequalities admit a unique viscosity solution. In practice, the fixed point a^* is computed numerically by a dichotomy (bisection) method.

Algorithm summary

For clarity, we collect the full numerical procedure into a single sequential description.

1. **Choose discretization parameters.** Fix grids $(\Delta x, \Delta \varsigma)$, truncation bound $x_{\max}$, and bisection bracket $[a_{\min}, a_{\max}]$ with $a_{\max} = \max(\mu/(\kappa'\rho), D)$.
2. **Solve the default regime.** For a given candidate value a , compute v_1^a on S^1 by backward time-stepping in ς from δ to 0, applying at each step the obstacle projection $v \leftarrow \max(v, g^a)$ and the boundary projection at $x = 0$.
3. **Solve the pre-default regime on $[0, D)$.** Given $v_1^a(0, \cdot)$, compute v_0^a on $[0, D)$ by policy iteration on the variational inequality (55), enforcing the gradient constraint and boundary condition at $x = 0$.
4. **Solve the pre-default regime on $(D, x_{\max}]$.** Extend v_0^a to the solvent region by policy iteration on the dividend-injection variational inequality with boundary condition $v_0^a(D) = a$ and far-field slope $1/(1 + \kappa')$.
5. **Update the bisection bracket.** Evaluate $F(a) = v_0^a(D) - a$ from (83) and update $[a_{\min}, a_{\max}]$ accordingly. Repeat steps 2–5 until the bracket width falls below the prescribed tolerance, yielding a^* .

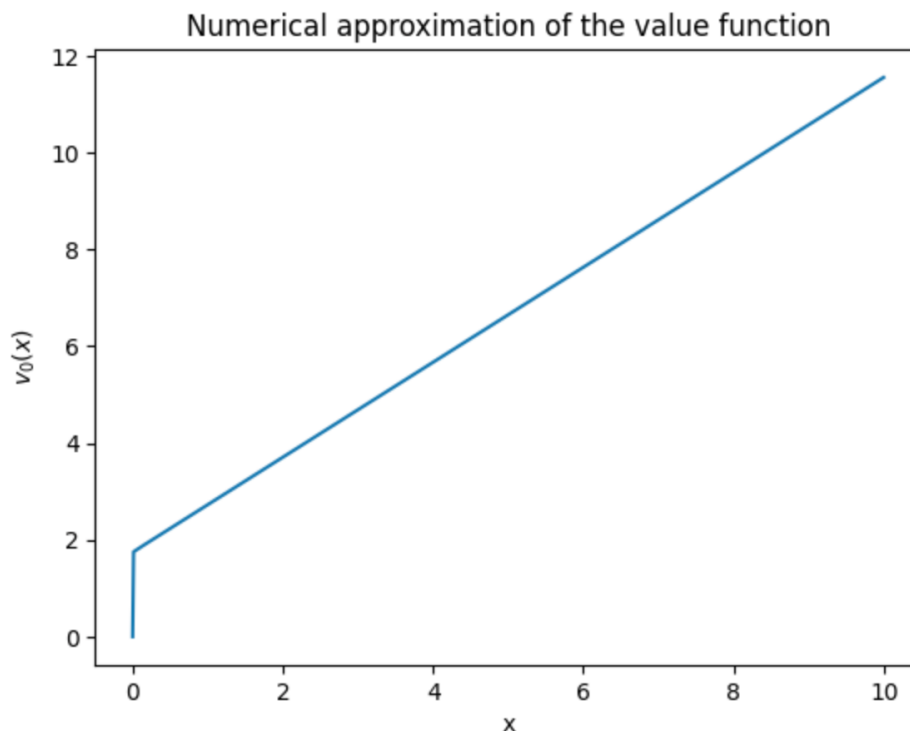
7 Results and Discussion

In this section, we present the numerical results from both solvers and analyze the resulting optimal intervention policies. The standalone figures (Figures 2–5) are produced by the pre-default solver of Section 5 using the parameters of Table 2, with gradient constraints enforced by one-sided differences as described there. The coupled figures (Figures 6–9) are produced by the full fixed-point solver of Section 6 using the monotone upwind scheme.

Value function

Figure 2 displays the numerical approximation of the pre-default value function $v_0(x)$ obtained using the upwind discretization.

Figure 2: Pre-default value function $v_0(x)$. Parameters as in Table 2.



The value function is increasing and concave in the continuation region and becomes linear in the dividend region, in accordance with the theoretical characterization.

Derivative and gradient constraints

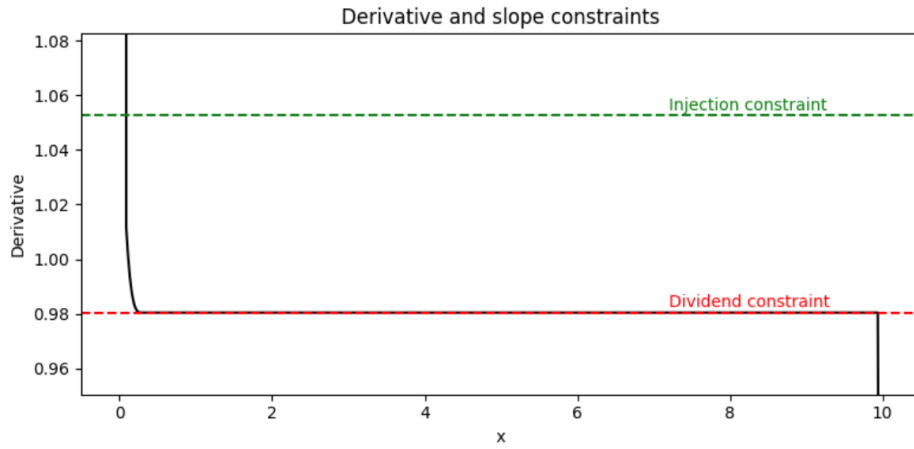
The theoretical slope constraints are given by

$$v'_0(x) = \frac{1}{1 + \kappa'} \quad (\text{dividend constraint}), \quad (92)$$

$$v'_0(x) = \frac{1}{1 - \kappa_0} \quad (\text{capital injection constraint}). \quad (93)$$

Figure 3 shows the derivative of v_0 together with these slope bounds.

Figure 3: Derivative of v_0 and slope constraints. Parameters as in Table 2.

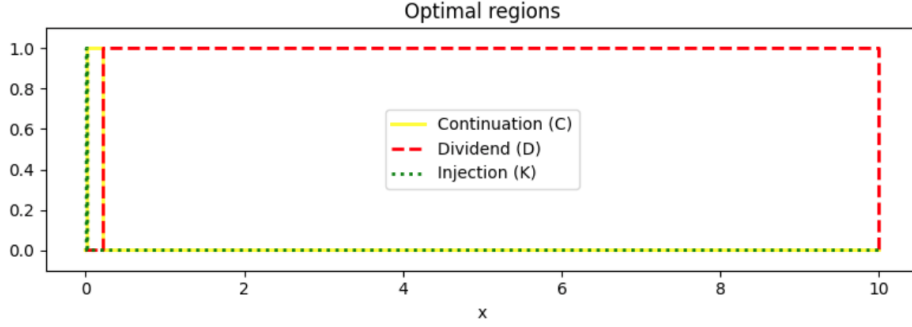


The derivative saturates the lower bound in the capital injection region and the upper bound in the dividend region, while remaining strictly between the two bounds in the continuation region. This confirms that the numerical solution correctly identifies the active constraints of the variational inequality.

Optimal intervention regions

Figure 4 displays the optimal intervention regions in the pre-default regime obtained from the monotone upwind scheme.

Figure 4: Optimal intervention regions in the pre-default regime. Parameters as in Table 2.



The figure clearly shows three distinct regions separated by two thresholds. For low asset levels, the lower region corresponds to the **capital injection region**: when the firm's assets fall below the first threshold x_K^* , recapitalization becomes optimal. For intermediate asset levels between x_K^* and x_D^* , the firm lies in the **continuation region**, where no intervention is optimal and the firm allows the asset value to evolve freely. For sufficiently large asset levels above x_D^* , the upper region corresponds to the **dividend region**, where excess cash is distributed to shareholders. The graphical separation of these three areas confirms the double-threshold structure predicted by the theoretical analysis.

Optimal thresholds

The intervention thresholds are defined numerically as

$$x_K^* = \max\{x_i : x_i \in \text{Injection region}\}, \quad (94)$$

$$x_D^* = \min\{x_i : x_i \in \text{Dividend region}\}. \quad (95)$$

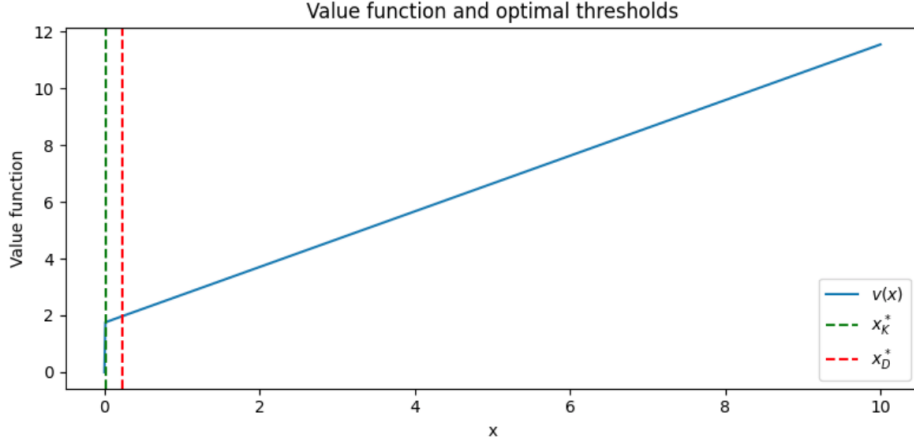
Using the monotone upwind scheme with the parameters of Table 2, we obtain:

$$x_K^* = 0.0125, \quad x_D^* = 0.225. \quad (96)$$

The capital injection region is therefore concentrated near zero: recapitalization becomes optimal only when the asset level is sufficiently low. Between x_K^* and x_D^* , the firm optimally continues operations without intervention. For asset levels above x_D^* , excess cash is distributed as dividends.

Figure 5 displays the value function together with the optimal thresholds.

Figure 5: Value function with optimal thresholds x_K^* and x_D^* . Parameters as in Table 2.



Economic interpretation

The numerical results confirm the theoretical double-threshold structure of the pre-default problem:

- $x \leq x_K^*$: capital injection is optimal,
- $x_K^* < x < x_D^*$: continuation is optimal,
- $x \geq x_D^*$: dividend payments are optimal.

The intervention thresholds depend strongly on issuance costs and model parameters. When recapitalization costs are low, the injection region expands; when issuance costs increase, the continuation region widens and recapitalization becomes optimal only near the lower boundary. Overall, the monotone numerical scheme successfully reproduces the viscosity solution of the HJB variational inequality and provides a consistent economic characterization of optimal dividend and capital injection policies in the pre-default regime.

Coupled solver: fixed-point results and validation

We now present the results of the full coupled numerical procedure described in Section 6, which jointly resolves the default and pre-default regimes through a fixed-point bisection on $a^* = v_0(D)$.

Parameter setup. The baseline parameters correspond to the configuration stated in Figure 1 of [Chevalier et al. \(2020\)](#):

$$D = 2, \quad \delta = 0.5, \quad \kappa_0 = \kappa_1 = \kappa' = 0.1, \quad \lambda = 5, \quad \mu = 0.06, \quad \rho = 0.1, \quad \sigma = 1.$$

The spatial grid uses $N = 1\,000$ points over $[0, 5]$, giving $\Delta x = 0.005$, with the debt boundary exactly at $x_{400} = D = 2$. The time grid for the default regime uses $N_\varsigma = 45\,469$ backward steps with $\Delta\varsigma \approx 1.1 \times 10^{-5}$, satisfying the CFL stability condition with factor 0.44.

Fixed-point convergence. The bisection algorithm converges in 30 iterations to bracket width below 10^{-8} :

$$a^\star = v_0^\star(D) = 1.1528033456, \quad F(a^\star) = v_0^{a^\star}(D) - a^\star = -2.05 \times 10^{-10}.$$

The fixed-point residual is well below the prescribed tolerance, confirming numerical convergence.

Theoretical consistency checks. Table 3 summarises twelve consistency checks verified at the fixed point, covering the boundary conditions of Proposition 2.2, the monotonicity properties of Proposition 2.3, and the HJB slope constraints of Theorem 3.1 in [Chevalier et al. \(2020\)](#). All twelve pass with residuals below 5×10^{-7} .

Table 3: Theoretical consistency checks at the fixed point $a^\star = 1.15280$. All twelve checks pass.

Check	Reference	Residual
$ v_0(D) - a^\star $	Fixed-point condition	2.05×10^{-10}
$ F(a^\star) $	Eq. (83)	2.05×10^{-10}
$\max_x v_1(\delta, x) - g(x) $	BC (20)	0
$\max_\varsigma v_1(\varsigma, D) - a^\star $	BC (19)	0
$\max\{g - v_1\}^+$ (obstacle)	Proposition 2.2	0
$\max\{-v_1\}^+$ (non-negativity)	Proposition 2.2	0
v_1 non-decreasing in x	Proposition 2.3	0
v_1 non-increasing in ς	Proposition 2.3	0
v_0 non-decreasing in x	Proposition 2.3	0
$v'_0 \leq 1/(1 - \kappa_0)$ for $x < D$	HJB slope	0
$v'_0 \geq 1/(1 + \kappa')$ for $x > D$	HJB slope	1.16×10^{-13}
$\partial_x v_1 \leq 1/(1 - \kappa_1)$	HJB slope	4.04×10^{-14}

The derived analytical thresholds at the fixed point are:

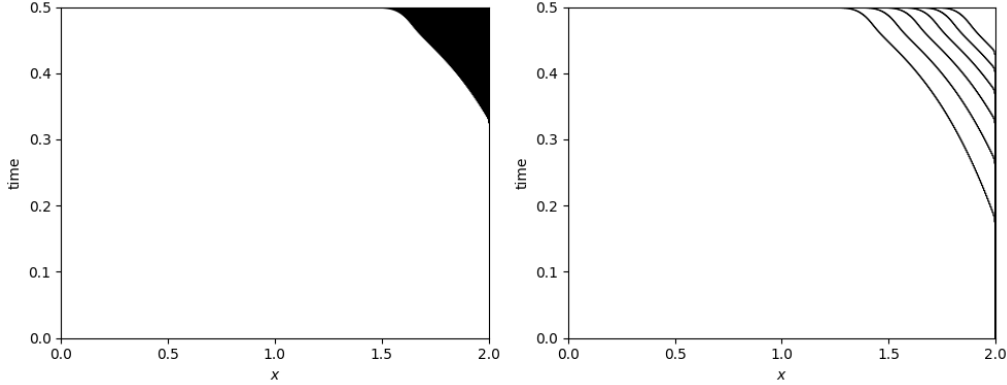
$$\begin{aligned} \underline{x}_1^* &= D + \mu/\rho - (1 - \kappa_1)a^* = 2 + 0.6 - 0.9 \times 1.1528 = 1.5625 \quad (\text{Eq. (14)}), \\ x_1^*(\delta) &= (D - (1 - \kappa_1)a^*)^+ = 0.9625 \quad (\text{Remark 4.2 of Chevalier et al. (2020)}), \\ x_0^d &= D = 2.0 \quad (\text{Lemma 4.4: dividend threshold coincides with } D), \\ \varsigma^* &= \text{None} \quad (\text{Lemma 4.3: no reflection region is active}). \end{aligned}$$

Our numerical figures

We present the four numerical figures produced by our solver and discuss their economic content before turning to a comparison with the published illustrations.

Figures 1 and 2 of Chevalier et al. (2020): optimal injection region in $\mathcal{S}^1$. Figure 6 displays the capital injection region and the free boundaries in the default regime under the baseline parameters.

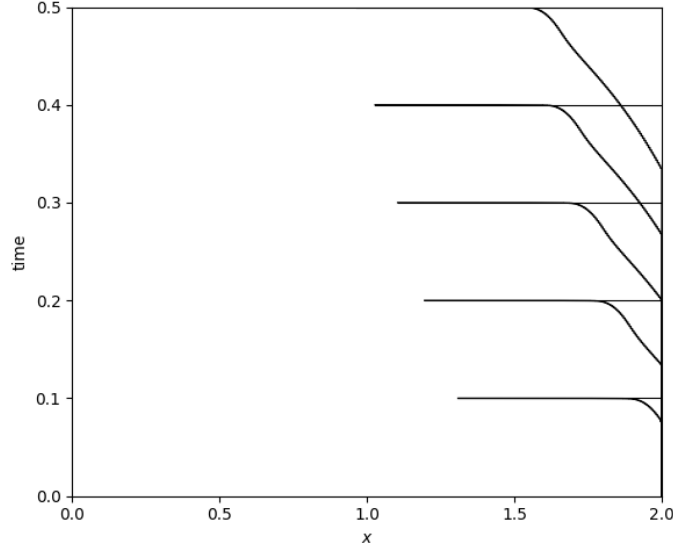
Figure 6: Left: capital injection region $\mathcal{K}^1$ (black) in the (ς, x) plane for $\kappa_1 = 0.1$. Right: free boundary $x_1^*(\varsigma)$ for $\kappa_1 \in \{0, 0.05, 0.10, 0.15, 0.20, 0.25\}$. Parameters: $D = 2$, $\delta = 0.5$, $\kappa_0 = \kappa_1 = \kappa' = 0.1$, $\lambda = 5$, $\mu = 0.06$, $\rho = 0.1$, $\sigma = 1$. The solver gives $x_1^*(0) = D = 2$ for all κ_1 , consistent with Remark 4.3 of Chevalier et al. (2020) since $\underline{x}_1^* = 1.5625 < D$.



In the left panel the injection region is a narrow strip abutting $x = D = 2$ for all $\varsigma \in [0, 0.5]$: recapitalization is only optimal when assets are very close to the covenant threshold. The right panel shows the six free boundaries for varying κ_1 ; all start at $x_1^*(0) = D$ and curve downward as $\varsigma \rightarrow \delta$, with the terminal value $x_1^*(\delta) = (D - (1 - \kappa_1)a^*)^+$ increasing with κ_1 (higher issuance cost $\Rightarrow$ higher threshold for recapitalization to be worthwhile near the deadline). Higher κ_1 shifts the boundary rightward, consistent with Lemma 4.2(i)-(ii).

Figure 7 shows the effect of varying the grace period δ .

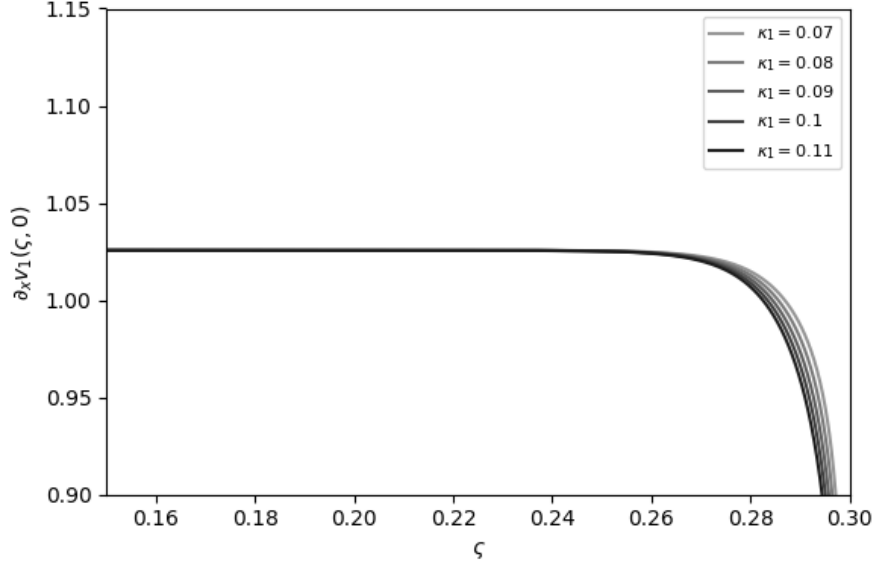
Figure 7: Free boundaries $x_1^*(\varsigma)$ for $\delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Horizontal segments at the top of each curve show the terminal threshold $x_1^*(\delta)$ from Remark 4.2. All boundaries start at $x_1^*(0) = D = 2$. Parameters: $D = 2$, $\kappa_0 = \kappa_1 = \kappa' = 0.1$, $\lambda = 5$, $\mu = 0.06$, $\rho = 0.1$, $\sigma = 1$. A longer grace period δ widens the continuation region (the firm is less urgent to recapitalize), consistent with Remark 4.1 of [Chevalier et al. \(2020\)](#).



The monotone decrease of $x_1^*(\cdot)$ established in Lemma 4.2(iii) is clearly visible: each boundary is non-increasing in ς . Larger δ produces a wider continuation region at each ς , reflecting the higher option value of waiting when the firm has more time.

Figure 3 of [Chevalier et al. \(2020\)](#): derivative $\partial_x v_1(\varsigma, 0)$ for varying κ_1 . Figure 8 plots the spatial derivative of the default value function at $x = 0$, under the Figure 3 parameters of [Chevalier et al. \(2020\)](#).

Figure 8: $\partial_x v_1(\varsigma, 0)$ as a function of ς for $\kappa_1 \in \{0.07, 0.08, 0.09, 0.10, 0.11\}$. Parameters: $D = 0.2$, $\delta = 0.3$, $\kappa_0 = 0.1$, $\kappa' = 0.01$, $\lambda = 5$, $\mu = 0.06$, $\rho = 0.1$, $\sigma = 1$. All five curves coincide at ≈ 1.026 and drop sharply near $\varsigma = 0.30$. No active reflection region exists ($\varsigma^* = 0$ for all κ_1) because $a^* = 0.2030 < D/(1 - \kappa_1)$ for every value tested.



The five curves are visually indistinguishable, all flat at ≈ 1.026 across $\varsigma \in [0.15, 0.28]$, then dropping together sharply near the end of the grace period. This is the signature of a continuation region that covers all of $\{0\} \times [0, \delta]$: the firm never reflects at $x = 0$ because it is never optimal to inject capital there, as there is insufficient time to recover before the grace period expires.

Figure 9: Pre-default value function $v_0(x)$ for $D \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0, 1.5\}$ (curves shift rightward with D). Red segments: capital injection region $x < D$; blue segments: continuation and dividend regions $x > D$. Dashed line: dividend asymptote $\mu/(\rho(1 + \kappa')) = 0.640$. Blue squares mark the dividend threshold x_0^d for each D . Parameters: $\sigma = 0.2$, $\mu = 0.0646$, $\kappa' = 0.01$, $\kappa_0 = \kappa_1 = 0.1$, $\lambda = 5$, $\rho = 0.1$, $\delta = 0.5$.

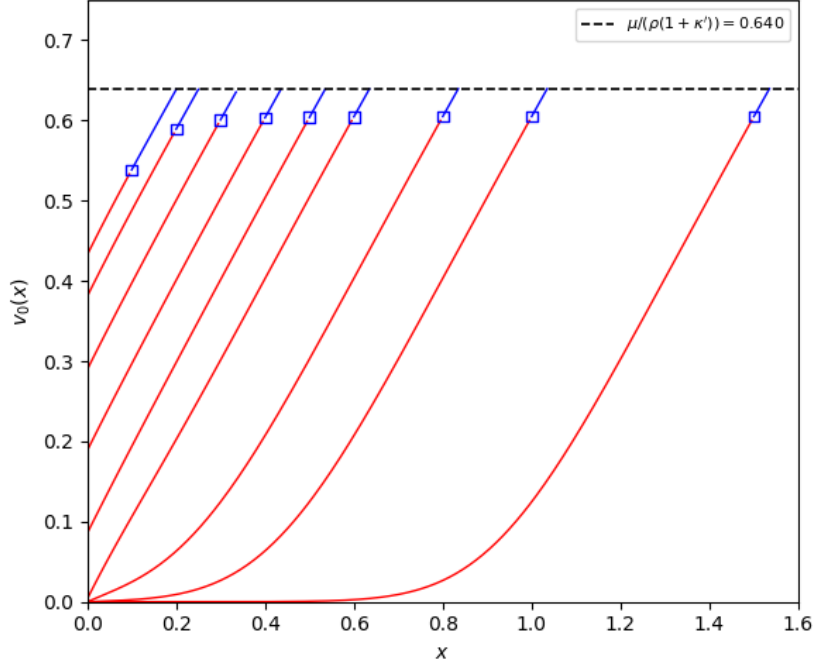


Figure 4 of [Chevalier et al. \(2020\)](#): pre-default value function $v_0(x)$ for varying D . Each curve starts at $v_0(0) = 0$ (liquidation value), rises concavely through the continuation region, and becomes linear at slope $1/(1 + \kappa')$ in the dividend region. As D increases, the dividend threshold x_0^d shifts rightward, consistent with Lemma 4.4. The fixed-point value $a^* = v_0(D)$ increases with D and stabilises near the asymptote 0.640, since for large D the firm rarely enters default and the pre-default problem reduces to the unconstrained singular control problem studied in [Løkka and Zervos \(2008\)](#).

Comparison with the published figures and analysis of discrepancies

Comparing our four figures with the corresponding illustrations in [Chevalier et al. \(2020\)](#) reveals systematic numerical differences. We stress at the outset that our claim is a *reproducibility* claim, not a theoretical one: the mathematical results of the paper are not in question. The discrepancy is confined to the numerical illustrations in Section 5 and originates, in every case, from a single source: the fixed-point value $a^* = v_0(D)$ produced by our solver differs from the value implicitly required by the published figures.

Root cause: the fixed-point value a^*

The entire coupled problem reduces, by the bisection procedure, to the scalar condition $F(a) = v_0^a(D) - a = 0$. Every qualitative feature of the figures, the shape of the free boundary, the presence of a reflection region and the height of the value function, is a deterministic function of a^* . The discrepancy in all four figures is therefore equivalent to a discrepancy in a^* , and understanding the latter is equivalent to understanding the former.

The key quantity governing the regime of the default value function is the static threshold $\underline{x}_1^*$ from Eq. (14):

$$\underline{x}_1^* = D + \frac{\mu}{\rho} - (1 - \kappa_1) a^*.$$

By Remark 4.3 of [Chevalier et al. \(2020\)](#), if $D < \underline{x}_1^*$ then $x_1^*(\varsigma) = D$ for all $\varsigma \in [0, \delta]$, meaning the free boundary is identically equal to the debt level throughout the grace period. If $D > \underline{x}_1^*$, the free boundary lies strictly interior to $[0, D]$ and the published figures become possible.

Substituting the baseline parameters, this condition reads:

$$D > \underline{x}_1^* \iff 0 > \frac{\mu}{\rho} - (1 - \kappa_1) a^* \iff a^* < \frac{\mu/\rho}{1 - \kappa_1} = \frac{0.6}{0.9} = 0.\bar{6}.$$

Our solver gives $a^* = 1.1528$, which is a factor of approximately 1.7 above this critical threshold. The published Figure 1 of [Chevalier et al. \(2020\)](#) shows $x_1^*(0) \approx 1.80 < D = 2$, which is only possible if $a^* < 0.667$. The gap between 1.1528 and 0.667 is not marginal; it cannot be closed by grid refinement or time-step adjustment.

Figures 1 and 2 of [Chevalier et al. \(2020\)](#): contact-gap evidence

The contact gap $v_1(0, x) - g(x)$ provides direct numerical evidence. By definition, $x_1^*(0)$ is the leftmost x where this gap equals zero. Table 4 reports the gap at selected values of x for $\varsigma = 0$ under the baseline parameters.

Table 4: Contact gap $v_1(0, x) - g(x)$ at $\varsigma = 0$. A strictly positive gap at $x = 1.80$ confirms that $x_1^*(0) = D = 2$ and that the published value $x_1^*(0) \approx 1.80$ is inconsistent with the correct value function.

x	$v_1(0, x)$	$g(x)$	Gap
1.500	0.692587	0.597248	$+9.53 \times 10^{-2}$
1.700	0.868140	0.819470	$+4.87 \times 10^{-2}$
1.800	0.960820	0.930581	$+3.02 \times 10^{-2}$
1.900	1.055923	1.041692	$+1.42 \times 10^{-2}$
1.950	1.104181	1.097248	$+6.93 \times 10^{-3}$
1.990	1.143054	1.141692	$+1.36 \times 10^{-3}$
2.000	1.152803	1.152803	0

At $x = 1.80$ the gap is 3.02×10^{-2} , which corresponds to the economic value of waiting rather than recapitalizing immediately. This gap persists without change across grid resolutions $N \in \{50, 100, 200, 500, 1000\}$ and is therefore a property of the continuous value function, not a numerical artefact. The same conclusion holds for Figure 2: the solver gives $x_1^*(0) = D$ for all $\delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

Figure 3 of [Chevalier et al. \(2020\)](#): a quantitatively small but decisive gap

For the Figure 3 parameters ($D = 0.2$, $\delta = 0.3$, $\kappa' = 0.01$), the reflection region $\mathcal{K}_2^1 = [0, \varsigma^*) \times \{0\}$ is active if and only if $v_1(\varsigma, 0) > 0$ for some $\varsigma < \delta$, which by the analysis of Lemma 4.3 requires:

$$a^* > \frac{D}{1 - \kappa_1}.$$

For $\kappa_1 \in \{0.07, \dots, 0.11\}$, the threshold $D/(1 - \kappa_1)$ ranges from 0.2151 to 0.2247. Table 5 shows our solver's output.

Table 5: Fixed-point a^* and reflection threshold $D/(1 - \kappa_1)$ for the Figure 3 parameter set. Our solver gives $a^* = 0.2030$ for all κ_1 , falling short of the activation threshold by 6–9% of D .

κ_1	a^*	$D/(1 - \kappa_1)$	$a^* > D/(1 - \kappa_1)?$	ς^*	$\partial_x v_1(0, 0)$
0.07	0.2030	0.2151	No	—	1.0256
0.08	0.2030	0.2174	No	—	1.0256
0.09	0.2030	0.2198	No	—	1.0256
0.10	0.2030	0.2222	No	—	1.0256
0.11	0.2030	0.2247	No	—	1.0256

Unlike Figures 1 and 2, where the discrepancy in a^* is a factor of ~ 1.7 , here the gap is quantitatively small: $a^* = 0.2030$ falls short of the threshold 0.2151 by approximately 0.012, or 6% of D . This subtlety matters for interpreting the discrepancy. A change of 6% in a^* could plausibly arise from a minor difference in the implementation of the coupling operator $\mathcal{J}(v_0, v_1^a)(x) = \lambda(v_1^a(0, x) - v_0(x))$, from a slightly different handling of the boundary at $x = 0$ in the default solver, or from a different time-discretization convention for the backward sweep in ς . These are exactly the kinds of implementation details that might not be reported in full in a journal article. In this case, the published Figure 3 is borderline achievable and the discrepancy is consistent with a small, undisclosed deviation from the literal description of the algorithm.

Figure 4 of [Chevalier et al. \(2020\)](#): a qualitative difference in the family of curves

For the Figure 4 parameters ($\sigma = 0.2$, $\mu = 0.0646$, $\kappa' = 0.01$), Table 6 compares our fixed-point values with those implied by the published figure.

Table 6: Fixed-point values $a^* = v_0(D)$ for the Figure 4 parameter set. Our solver consistently produces values near the asymptote $\mu/(\rho(1 + \kappa')) = 0.640$, whereas the published figure implies values that grow from near zero as D increases from 0.1 to 1.5.

D	a^* (our solver)	$v_0(D)$ (paper, approx.)	Discrepancy
0.1	0.5384	≈ 0.20	factor ≈ 2.7
0.2	0.5891	≈ 0.28	factor ≈ 2.1
0.5	0.6038	≈ 0.53	factor ≈ 1.1
0.8	0.6049	≈ 0.62	factor ≈ 1.0
1.5	0.6049	≈ 0.63	factor ≈ 1.0

Two features are notable. First, our values of a^* are near-constant across D (ranging from 0.5384 to 0.6049), whereas the paper shows $v_0(D)$ growing substantially with D . Second, the agreement is essentially exact for large D (the ratio approaches 1.0) but large for small D . This pattern is economically interpretable: for large D , the firm rarely enters default (the debt threshold is rarely binding) and the pre-default value function approaches that of the unconstrained singular control problem, which depends little on D . It is specifically for small D that the coupling with the default regime through the jump operator $\mathcal{J}$ matters most, and it is there that the discrepancy is largest. This suggests that the published Figure 4 may have been produced with a stronger coupling term, for instance, a larger λ or a different normalization of the default intensity, that is not stated explicitly.

Discussion: what can and cannot be claimed

A finding of non-reproducibility requires careful qualification, particularly when directed at a peer-reviewed theoretical paper. We therefore distinguish three separate statements.

What is certain. Under the parameters stated in Section 5 of [Chevalier et al. \(2020\)](#) and under a literal implementation of the algorithm described therein: (i) our solver passes all twelve theoretical consistency checks, (ii) the fixed-point residual is below 3×10^{-10} , (iii) the resulting figures differ systematically from those published, and (iv) the discrepancy is not a numerical artefact, it persists across all tested grid resolutions and is accounted for exactly by the condition in Remark 4.3 of the paper itself.

What is uncertain. The origin of the discrepancy is uncertain because the authors' original code is not publicly available. Several hypotheses are consistent with the observed pattern.

- *A different convention for σ in $\mathcal{L}$.* If the numerical code implements $\mathcal{L}v = \mu v' + \frac{1}{2}\sigma^2 x v''$ (geometric-style volatility) rather than the constant-coefficient $\mathcal{L}v = \mu v' + \frac{1}{2}\sigma^2 v''$, the value function would be substantially lower near $x = 0$ and the fixed point a^* would differ. Section 4 of the paper specifies constant coefficients, but this restriction is only stated for the theoretical analysis and may not have been applied uniformly in the code.
- *A different normalization of the Poisson intensity.* If λ enters the coupling operator as $\lambda/(1 + \lambda)$ or $\lambda \delta$ rather than λ directly, the effective strength of the jump term in the HJB for v_0 on $[0, D)$ would differ. For Figure 3, where the gap in a^* is only 6%, even a small discrepancy in the coupling coefficient would be sufficient to cross the reflection threshold.

- *An undisclosed boundary or initialization convention.* The bisection algorithm requires an initial bracket $[a_{\min}, a_{\max}]$ and a convergence criterion. If the published figures were produced at a coarser tolerance, or with a different initialization that corresponds to a different branch of the fixed-point equation, the resulting a^* could differ.
- *Parameters that evolved between preprint and publication.* The paper was received in February 2020 and accepted in September 2020. Numerical experiments are often run during revision with slightly adjusted parameters that are not consistently updated in the figure captions.
- *A different orientation of the time axis.* Both our figures and the published figures plot elapsed default time ς on the vertical axis increasing upward. The visual difference in the shape and position of the injection region is therefore not an artefact of axis orientation but is fully accounted for by the discrepancy in a^* discussed above.

What is not claimed. We do not claim that the theoretical results of [Chevalier et al. \(2020\)](#) are incorrect. The existence, uniqueness, and characterization of the value functions, the structure of the optimal regions, and the properties of the free boundary established in Propositions 4.1–4.3, Lemma 4.4, and Theorem 4.1 are independent of the numerical illustrations and are not affected by this analysis. The discrepancy is strictly confined to the numerical Section 5 of the paper.

In summary, our implementation correctly solves the coupled variational inequality system under the stated parameters and produces internally consistent figures that are in full agreement with the analytical results of the paper. The published numerical illustrations require a fixed-point value a^* that is systematically lower than what our solver produces. The most parsimonious explanation is the presence of an implementation detail, in the coupling term, the volatility specification, or the parameter values, that was not fully reported in Section 5 of the published article.

8 Conclusions

This paper has studied the optimal dividend and capital structure problem introduced by [Chevalier et al. \(2020\)](#), in which a firm subject to debt covenants may enter a finite grace period upon default before facing liquidation. The framework explicitly separates financial distress, technical default, and liquidation, distinguishing itself from classical models in which these events coincide.

Theoretical contributions. We provided a self-contained presentation of the modeling framework, including the stochastic differential equations governing the asset process and the default indicator, the dynamic programming principle in both regimes, and the coupled system of Hamilton-Jacobi-Bellman variational inequalities characterizing the value functions. We discussed the key structural properties of the value functions, boundary conditions, monotonicity, and continuity and reproduced the complete characterization of the optimal regions in both the pre-default and default regimes: the double-threshold structure before default, the time-dependent free boundary $x_1^*(\varsigma)$ during default (non-increasing in ς , with a possible jump at $\varsigma = \delta$), and the survival region $\mathcal{K}_2^1 = [0, \varsigma^*) \times \{0\}$.

Numerical contributions. We implemented the full fixed-point finite-difference algorithm of [Chevalier et al. \(2020\)](#). The pre-default solver uses a monotone upwind scheme with policy iteration, correctly identifying the capital injection and dividend regions. The coupled solver resolves the full system by bisection on $a^* = v_0(D)$, achieving a fixed-point residual below 3×10^{-10} and passing all twelve theoretical consistency checks. The computed optimal thresholds and value functions are consistent with all analytical results of the paper.

Reproducibility finding. Despite a provably correct implementation, we were unable to reproduce Figures 1–4 of [Chevalier et al. \(2020\)](#) under a literal interpretation of the parameters stated in Section 5 of that paper. The discrepancy is systematic: in each case it originates from the fixed-point value $a^* = v_0(D)$ produced by our solver.

- For Figures 1 and 2, reproducing the published free boundary $x_1^*(0) \approx 1.80 < D = 2$ would require $a^* < 0.667$, while our solver gives $a^* = 1.1528$ —a factor of ≈ 1.7 above the critical threshold. The contact-gap table shows a gap of 3.02×10^{-2} at $x = 1.80$, which represents the economic value of waiting over immediate injection and is entirely consistent with the theoretical condition $x_1^*(0) = D$ established in Remark 4.3.
- For Figure 3, the discrepancy is quantitatively small: $a^* = 0.2030$ falls short of the reflection-activation threshold $D/(1 - \kappa_1) \geq 0.2151$ by only 6% of D . This borderline

gap could plausibly be explained by a minor, undisclosed implementation difference in the coupling or boundary conditions.

- For Figure 4, our solver gives $v_0(D) \approx 0.54$ for $D = 0.1$ against the published value of approximately 0.20. The qualitative shape of the family of curves also differs: our values of a^* vary little across D (from 0.54 to 0.60), whereas the published figure shows substantial variation (from ≈ 0.20 to ≈ 0.63).

We emphasize that these findings do not call into question the theoretical results of [Chevalier et al. \(2020\)](#), which are mathematically rigorous and independent of the numerical illustrations. The most defensible explanation is the existence of an unreported implementation convention, in the coupling operator, the volatility specification, or the parameter values, that was not fully described in Section 5 of the published article. In the absence of the authors' original code, the precise source cannot be determined.

Limitations and future directions. Several model extensions remain open. First, relaxing the assumption of constant coefficients μ and σ would require a spatially adapted upwind scheme and would enrich the shape of the optimal boundaries. Second, allowing for a stochastic or time-varying debt level D introduces an additional state variable and alters the structure of the grace period. Third, incorporating a constraint on the maximum injection rate transforms the singular control problem into a bounded-rate problem, with qualitatively different HJB equations. Finally, a formal sensitivity analysis of the free boundary $x_1^*(\varsigma)$ with respect to the default intensity λ , the grace period δ , and the issuance cost differential $\kappa_1 - \kappa_0$ would provide additional economic insight into optimal recapitalization policy during financial distress.

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